AI-assisted accounting: controlled architecture comparison

HOLDOUT | COMPLETE

Run: holdout-v1 | Dataset: daybook-study-v1

Measured snapshot generated 2026-10-31T07:38:19.338Z. The PDF summarizes saved results only; exporting does not run models.

Scope	Measured state
Main episodes	1620 recorded / 1620 planned; 0 missing
Main split	holdout
Repeats per question	3
Verifier snapshot	holdout; complete; 80 planned candidates per method
Snapshot integrity	verified
Human readability score	Not evaluated. No total score out of 100 is computed.

How to read this report

This compares nine configurations of one accounting assistant. The ledger remains the source of financial numbers; retrieval supplies documentary context. Optional reviewers inspect candidates but do not replace deterministic accounting checks.

Structured task success requires the declared behavior, facts, scope, evidence and assumptions. It is a machine-checkable proxy, not proof that every sentence is accurate. The automated rubric has a maximum of 90; the remaining 10 points require human clarity review.

No winning architecture is selected here. Incomplete or development-only observations cannot establish a held-out improvement. Missing measurements remain N/A, and failed attempts remain in end-to-end denominators.

Methods and experimental controls

Route	Query capability	Evidence retrieval	Answer review
R0	Fixed financial tools	No document retrieval	Code checks
R1	Broader analytical tools	No document retrieval	Code checks
R2	Same broader tools	Lexical documents	Code checks
R3	Same broader tools	Managed citation selection	Code checks
R4	Same broader tools	Embeddings + pgvector + lexical	Code checks
R5	Same broader tools	R4 + Jev reranking	Code checks
R6	Wren semantic SQL engine	Same custom hybrid	Code checks
R7	Same broader tools	Same custom hybrid	Code + LLM critic
R8	Same broader tools	Same custom hybrid	Code + Jev verifier

The shared synthetic ledger, document corpus, question groups and scoring reference are frozen. Candidate providers receive permitted questions and evidence, not hidden gold answers. Sources and queries are restricted to the authorized workspace; journal proposals are never automatically posted.

The study uses a common generator across routes; R0 reproduces the current tool architecture under that study model. R7 uses a separate critic context with the same model weights. Separate calls do not make generator and reviewer errors statistically independent.

R3 incurs an additional managed generation/search request. Its citations select authorized original documents; generated summaries are discarded. Opaque internal chunks and rankings prevent a clean algorithm-only retrieval comparison. R6 measures the restricted Wren engine integration, not the complete Wren product.

Failed requests are retained. Provider calls have bounded execution deadlines and no automatic retry. Pacing and local queue dispatch contribute to harness wall latency. Costs are paid-equivalent usage estimates plus reservations for unknown usage, not invoices or evidence that a free-tier request was charged.

Per-family outcomes and scenario-group bootstrap intervals preserve paraphrase/repeat clusters. Intervals are omitted when too few independent groups are available. Conditional reviewer metrics exclude provider failures; operational metrics block on failures and show that distinction.

Nine-route measured comparison

This holdout snapshot contains 1620/1620 planned episodes. Rows remain in predeclared route order. Provider errors and blocked answers remain in task-success denominators; missing episodes are disclosed rather than replaced.

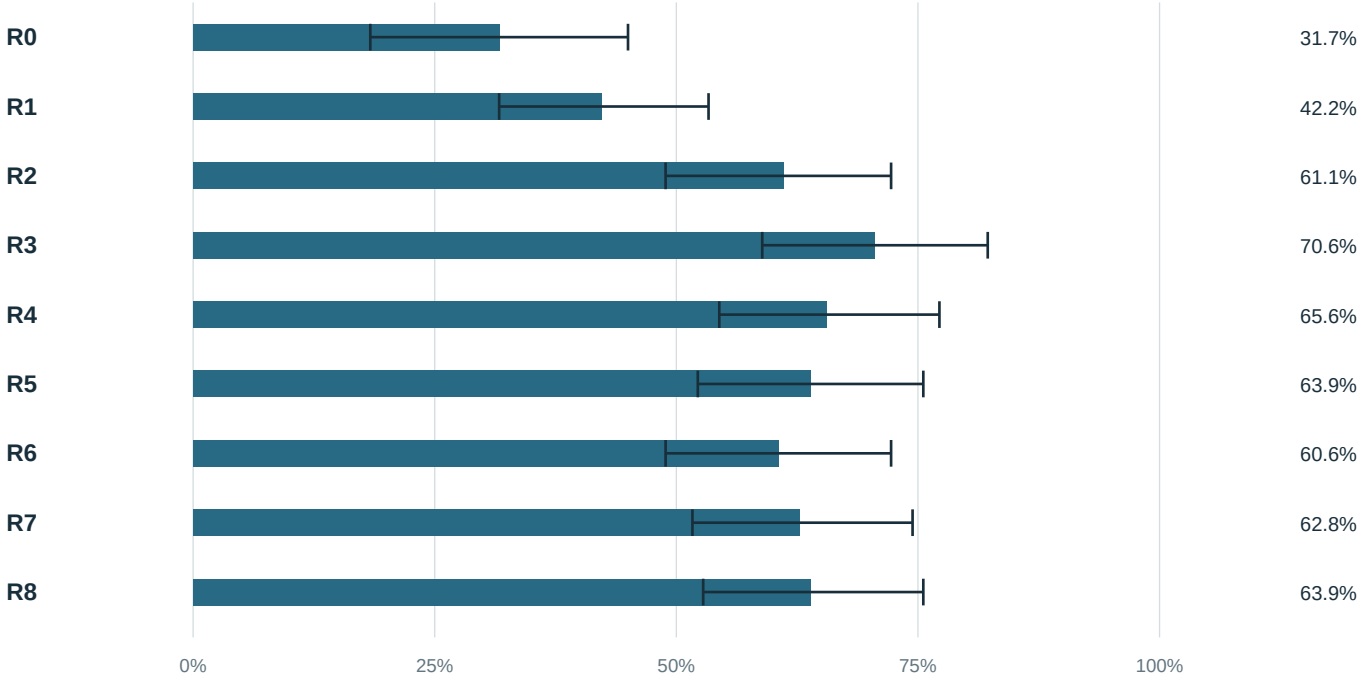
The all-family macro score gives equal weight to each of the six task families and stays N/A until every family has observations. The separate observed micro rate is useful for inspecting a partial run, but may change as later families are evaluated.

Route	n / planned	Micro success	All-family macro	Errors	Score / 90	Cost / episode
R0	180/180	31.7%	31.7%	2.8%	42.55	$0.0015
R1	180/180	42.2%	42.2%	0.6%	45.88	$0.0025
R2	180/180	61.1%	61.1%	0.6%	58.80	$0.0030
R3	180/180	70.6%	70.6%	1.7%	65.08	$0.0042
R4	180/180	65.6%	65.6%	0.6%	62.23	$0.0031
R5	180/180	63.9%	63.9%	0.0%	59.89	$0.0032
R6	180/180	60.6%	60.6%	2.8%	57.14	$0.0029
R7	180/180	62.8%	62.8%	1.7%	59.34	$0.0035
R8	180/180	63.9%	63.9%	0.6%	61.01	$0.0032

Micro success uses all recorded episodes. All-family macro is N/A until every task family is represented. Recorded-error rate includes transport/provider failures, not every blocked answer.

Task success across families

Figure 1. All-family task success and recorded 95% interval



Bars require all-family coverage. Whiskers reproduce the saved 95% scenario-group bootstrap interval; no interval is manufactured for a sparse pilot. N/A does not mean zero.

Figure 2. Structured task success within each task family

	financial	analytics	documents	scenario	entry	adversarial
R0	60% n=30	0% n=30	0% n=30	20% n=30	70% n=30	40% n=30
R1	60% n=30	37% n=30	0% n=30	17% n=30	73% n=30	67% n=30
R2	63% n=30	40% n=30	77% n=30	33% n=30	83% n=30	70% n=30
R3	73% n=30	53% n=30	90% n=30	40% n=30	90% n=30	77% n=30
R4	70% n=30	43% n=30	80% n=30	37% n=30	90% n=30	73% n=30
R5	73% n=30	43% n=30	80% n=30	33% n=30	87% n=30	67% n=30
R6	40% n=30	33% n=30	80% n=30	40% n=30	93% n=30	77% n=30
R7	60% n=30	40% n=30	87% n=30	37% n=30	87% n=30	67% n=30
R8	57% n=30	40% n=30	87% n=30	43% n=30	87% n=30	70% n=30

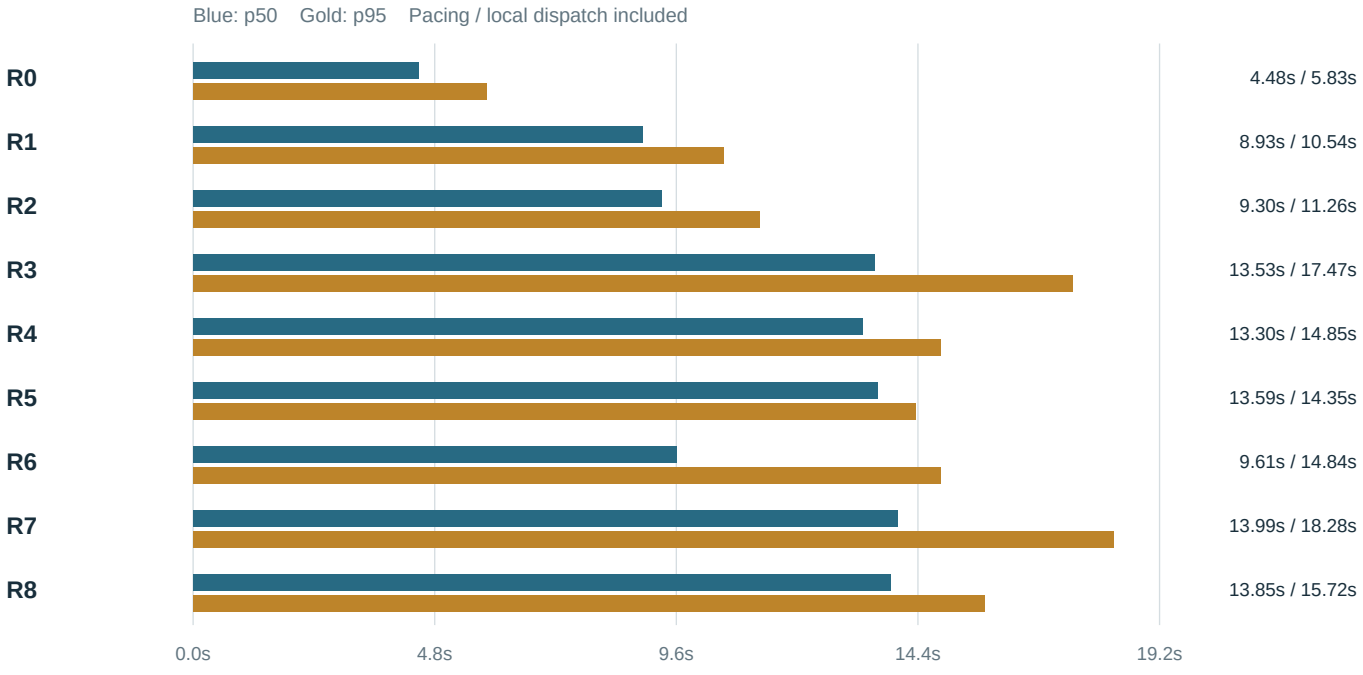
Each cell shows measured success and its episode denominator. Missing cells are N/A. Families have equal weight in the all-family macro score, despite possible unequal recorded counts during a partial run.

Cost and latency tradeoffs

Figure 3. Cost per recorded episode versus task success

The scatter uses observed micro success so partial data remain visible. It is descriptive, not a Pareto winner selection. Red marks indicate a recorded hard-gate disqualification. Missing cost or outcome values are omitted.

Figure 4. Harness wall latency: median and 95th percentile



Wall time includes provider calls, retrieval, validation, rate pacing and local dispatch delays. It does not establish production serving latency or million-user capacity.

Latency denominators and provider service time

Complete HTTP totals require measured service durations for every recorded provider attempt in an episode. Partial known subtotals are shown separately. Pacing and queue waits remain in wall time. These measurements are neither single-model latency nor production request latency. Non-error outcomes include blocks.

Route	Scope	n	Wall p50 / p95	Complete HTTP p50 / p95	HTTP C/P/U/N	Partial subtotal p50 / p95
R0	All	180	4.48s / 5.83s	1.42s / 2.73s	179/0/1/0	N/A / N/A
R0	Non-error	175	4.47s / 5.83s	1.42s / 2.73s	175/0/0/0	N/A / N/A
R1	All	180	8.93s / 10.54s	2.75s / 4.59s	179/0/1/0	N/A / N/A
R1	Non-error	179	8.93s / 10.54s	2.75s / 4.59s	179/0/0/0	N/A / N/A
R2	All	180	9.30s / 11.26s	2.89s / 5.19s	179/0/0/1	N/A / N/A
R2	Non-error	179	9.30s / 11.27s	2.89s / 5.19s	179/0/0/0	N/A / N/A
R3	All	180	13.53s / 17.47s	6.64s / 10.82s	180/0/0/0	N/A / N/A
R3	Non-error	177	13.53s / 17.73s	6.64s / 11.35s	177/0/0/0	N/A / N/A
R4	All	180	13.30s / 14.85s	3.12s / 4.78s	180/0/0/0	N/A / N/A
R4	Non-error	179	13.32s / 14.86s	3.12s / 4.78s	179/0/0/0	N/A / N/A
R5	All	180	13.59s / 14.35s	3.27s / 5.01s	180/0/0/0	N/A / N/A
R5	Non-error	180	13.59s / 14.35s	3.27s / 5.01s	180/0/0/0	N/A / N/A
R6	All	180	9.61s / 14.84s	2.81s / 4.50s	180/0/0/0	N/A / N/A
R6	Non-error	175	9.62s / 14.84s	2.81s / 4.50s	175/0/0/0	N/A / N/A
R7	All	180	13.99s / 18.28s	4.00s / 5.91s	179/0/1/0	N/A / N/A
R7	Non-error	177	13.99s / 18.28s	4.00s / 5.91s	177/0/0/0	N/A / N/A
R8	All	180	13.85s / 15.72s	3.24s / 4.94s	180/0/0/0	N/A / N/A
R8	Non-error	179	13.93s / 15.78s	3.24s / 5.04s	179/0/0/0	N/A / N/A

C/P/U/N = complete / partial / unknown-or-unverified / no recorded calls. Complete-total percentiles exclude incomplete and no-call episodes. Partial-subtotal percentiles include only episodes with some measured calls and some missing timings. Missing durations never become zero. Legacy episode aggregates are unverified without per-call records; a timeout aggregate of 0 is not instantaneous HTTP execution.

Paired comparisons and cost accounting

Test	Comparison	Paired effect	Recorded 95% CI	Pairs	Unpaired T / C
H1	R1 - R0	10.6%	-4.4% to 25.6%	180	0 / 0
H2	R2 - R1	18.9%	10.6% to 27.8%	180	0 / 0
H3a	R3 - R2	9.4%	3.3% to 15.6%	180	0 / 0
H3b	R3 - R4	5.0%	0.0% to 10.6%	180	0 / 0
H4	R4 - R2	4.4%	0.6% to 8.3%	180	0 / 0
H5	R5 - R4	-1.7%	-6.1% to 2.2%	180	0 / 0
H6	R6 - R4	-5.0%	-12.8% to 2.2%	180	0 / 0
H7a	R7 - R4	-2.8%	-6.1% to 0.6%	180	0 / 0
H7b	R8 - R4	-1.7%	-5.6% to 1.7%	180	0 / 0

Effects are absolute percentage-point differences, macro-averaged over represented paired families. Positive favors the treatment. Sparse/incomplete comparisons do not cover every planned family; missing pairs and intervals are explicit. These multiple exploratory comparisons are not corrected into a single confirmatory significance claim.

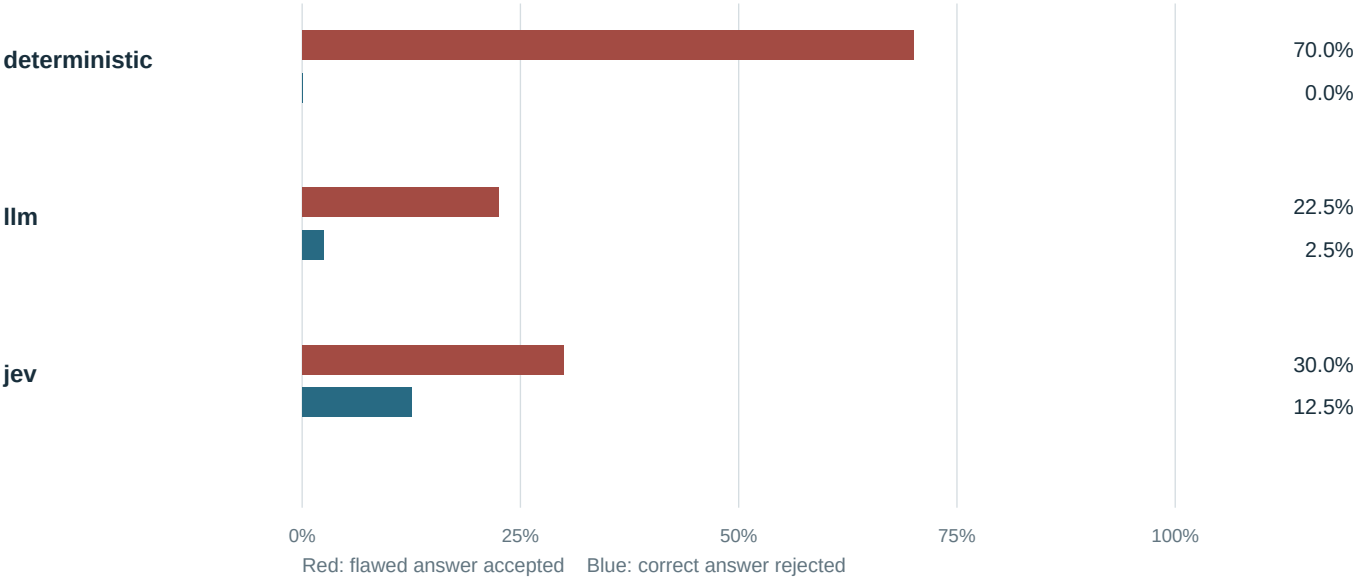
Route	Reported estimate	Unknown reservation	Unknown calls	Budget / success
R0	$0.2707	$0.0077	1	$0.0049
R1	$0.4366	$0.0094	1	$0.0059
R2	$0.5402	$0.0000	0	$0.0049
R3	$0.7592	$0.0000	0	$0.0060
R4	$0.5215	$0.0358	107	$0.0047
R5	$0.5369	$0.0361	108	$0.0050
R6	$0.5014	$0.0270	81	$0.0048
R7	$0.5951	$0.0429	101	$0.0056
R8	$0.5319	$0.0354	106	$0.0049

Setup/indexing budget: $0.0698. Unassigned run usage: $0.0000. A cost per success is N/A when there are no successes. Reservations retain uncertainty instead of declaring failed requests free.

Validator stress study - HOLDOUT

40 correct and 40 deliberately flawed candidate answers. Jev threshold: 0.8. Snapshot: COMPLETE.

Figure 5. Conditional reviewer errors on completed decisions



Reviewer	n / plan	Errors	Conditional FA / FR	Operational FA / FR	With code FA / FR
deterministic	80/80	0	70.0% / 0.0%	70.0% / 0.0%	70.0% / 0.0%
llm	80/80	0	22.5% / 2.5%	22.5% / 2.5%	17.5% / 2.5%
jev	80/80	0	30.0% / 12.5%	30.0% / 12.5%	15.0% / 12.5%

FA: flawed answer accepted. FR: correct answer rejected. Conditional denominators include completed reviewer judgments only. Operational metrics include provider failures as blocked outcomes; that is not proof that the reviewer detected the flaw. The combined column adds the deterministic checker.

Threshold selection uses development labels only, with a predeclared additional false-rejection constraint. Development results include threshold tuning and are not an unbiased held-out estimate. Maximum Jev hazard score is not a calibrated probability that the whole answer is wrong; no overall Brier score or human-judged truth score is invented.

These stress cases have 50% flawed prevalence and repeat source templates. They evaluate supplied candidate/evidence pairs, not production fraud prevalence, exhaustive search completeness, or independent human adjudication.

Supplemental retrieval and failure analysis

Conditional retrieval metrics include only document-applicable episodes with recorded completed retrieval. Unknown error stages and planner-skipped searches are counted separately; they are not silently scored as empty retrieval.

Route	Complete / eligible	Selected coverage	Recall@6	MRR@6	nDCG@6	E2E docs
R0	0/66	N/A	N/A	N/A	N/A	0.0%
R1	0/66	N/A	N/A	N/A	N/A	0.0%
R2	63/66	100.0%	100.0%	0.976	0.972	63.6%
R3	61/66	97.5%	N/A	N/A	N/A	68.2%
R4	61/66	98.4%	98.4%	0.975	0.962	63.6%
R5	63/66	98.4%	98.4%	1.000	0.985	66.7%
R6	62/66	99.2%	99.2%	0.992	0.980	71.2%
R7	59/66	99.2%	99.2%	0.992	0.979	65.2%
R8	63/66	98.4%	98.4%	0.992	0.977	68.2%

R3 managed citation selection has observable selected-document coverage; its raw search ranking metrics remain N/A. End-to-end document availability includes errors and blocks in the denominator, but does not imply answer correctness. Relevance uses exact required document/version IDs with binary gains.

Most frequent observed conditions

Condition	Episodes
structured_answer:required_fact_mismatch	540
structured_answer:wrong_behavior	446
structured_answer:required_fact_missing_or_multiple_mismatches	431
structured_answer:missing_ledger_provenance	420
structured_answer:missing_document_evidence	282
gate:deterministic_rejection	262
structured_answer:deterministic_validation_failed	262
structured_answer:missing_assumption	219
capability:documents_unavailable	132
structured_answer:unsupported_or_unassessed_fact	121
structured_answer:metric_key_mismatch	120
structured_answer:missing_clarification_topic	49
structured_answer:proposal_line_mismatch	27
planner:retrieval_not_requested	23
structured_answer:episode_error	20

Conditions overlap and are not proven root causes. Key/schema/scope mismatch is not automatically an arithmetic error. Full per-episode records remain in the supplemental JSON.

Shared capabilities: a secondary comparison

12 questions in 6 scenario groups. Eligibility: complete measurement.

Membership was defined from available tools before held-out results. This subset does not replace the six-family primary score; whole-benchmark hard gates still apply.

Route	Observed / planned	Micro success	Observed-family macro	Errors	Hard gate
R0	36/36	91.7%	91.7%	3	No observed violation
R1	36/36	72.2%	72.2%	0	No observed violation
R2	36/36	63.9%	63.9%	0	No observed violation
R3	36/36	75.0%	75.0%	0	No observed violation
R4	36/36	72.2%	72.2%	0	No observed violation
R5	36/36	72.2%	72.2%	0	No observed violation
R6	36/36	69.4%	69.4%	1	No observed violation
R7	36/36	66.7%	66.7%	0	No observed violation
R8	36/36	66.7%	66.7%	0	No observed violation

The held-out subgroup has only six scenario groups and covers financial and entry tasks. Paired intervals and exact membership are in the accompanying common-capability analysis. No additional winner is chosen from this subgroup.

Sensitivity to fixed rubric weights

All five columns are automated scores out of 90, equally averaged across task families. The human clarity dimension remains unavailable. These weights were declared during development; no favorable column is selected as the result.

Route	Original	Financial emphasis	Evidence emphasis	Uncertainty emphasis	Equal dimensions	Rank range
R0	42.55	41.30	46.22	47.60	47.40	8 to 9
R1	45.88	45.87	45.83	46.73	46.61	8 to 9
R2	58.80	58.50	59.30	59.62	59.71	6 to 6
R3	65.08	64.92	65.44	65.72	65.71	1 to 1
R4	62.23	61.90	62.71	63.24	63.34	2 to 2
R5	59.89	59.65	60.29	60.67	60.72	4 to 4
R6	57.14	56.89	57.61	58.17	58.17	7 to 7
R7	59.34	59.21	59.75	60.18	60.10	5 to 5
R8	61.01	60.75	61.45	62.01	62.04	3 to 3

Weighting	Factual	Completion	Grounding	Uncertainty
Original	40	20	20	10
Financial emphasis	50	15	15	10
Evidence emphasis	30	15	35	10
Uncertainty emphasis	30	15	20	25
Equal dimensions	22.5	22.5	22.5	22.5

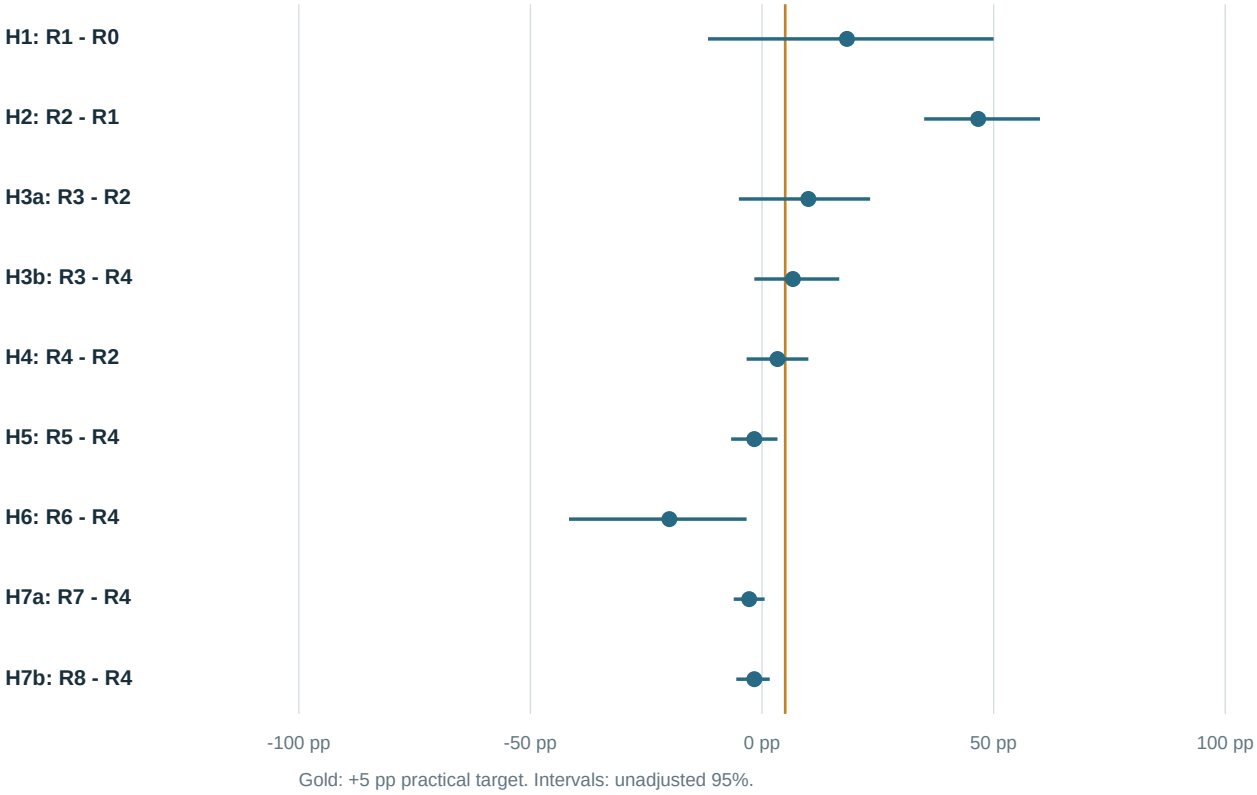
Errors retain their zero rubric components; missing observations remain missing. Rank ranges are unavailable when the analysis is ineligible. Stable ranks do not establish prose faithfulness or statistical superiority.

Predeclared component decision criteria

These targeted comparisons supplement the primary six-family result. Financial and analytics tasks assess broader tools and Wren; document and scenario tasks assess retrieval and reranking. The practical target is +5 percentage points. Uncertain tradeoffs remain inconclusive.

Interpretation eligibility: verified

Targeted component effects: paired structured success



Gold: +5 pp practical target. Intervals: unadjusted 95%.

Each point is a treatment-minus-control difference in structured success. Scenario-group resampling keeps paraphrases and repeats together. Missing intervals are labeled CI N/A, not drawn as zero-width certainty. The few groups and multiple comparisons limit precision; the figure is not a significance claim.

Component assessments

Test	Route / control	Target families	Effect [interval]	Assessment
H1	R1 / R0	financial, analytics	+18.3 pp [-11.7, +50.0]	inconclusive
H2	R2 / R1	documents, scenario	+46.7 pp [+35.0, +60.0]	candidate for further experiment
H3a	R3 / R2	documents, scenario	+10.0 pp [-5.0, +23.3]	inconclusive
H3b	R3 / R4	documents, scenario	+6.7 pp [-1.7, +16.7]	inconclusive
H4	R4 / R2	documents, scenario	+3.3 pp [-3.3, +10.0]	inconclusive
H5	R5 / R4	documents, scenario	-1.7 pp [-6.7, +3.3]	below practical target
H6	R6 / R4	financial, analytics	-20.0 pp [-41.7, -3.3]	below practical target
H7a	R7 / R4	financial, analytics, documents, scenario, entry, adversarial	-2.8 pp [-6.1, +0.6]	inconclusive
H7b	R8 / R4	financial, analytics, documents, scenario, entry, adversarial	-1.7 pp [-5.6, +1.7]	inconclusive

A candidate for further experiment is not a production recommendation. No combination of components is assumed to inherit individually measured benefits. Source integrity, hard gates, provider errors, useful task completion, latency and cost still constrain the decision.

Reviewer tradeoffs against deterministic checks

Method	Flawed acceptance difference	Added correct withholding	Conditional flawed acceptance	Assessment
llm	-52.5 pp [-68.4, -35.0]	+2.5 pp [+0.0, +7.9]	-52.5 pp [-68.4, -35.0]	inconclusive
jev	-55.0 pp [-73.2, -35.7]	+12.5 pp [+0.0, +31.6]	-55.0 pp [-73.2, -35.7]	below practical target

The point criterion requires less operational acceptance of flawed candidates and no more than +5 pp additional withholding of correct candidates. Conditional comparisons retain completed paired judgments only. Provider errors are withholding, not evidence that a flaw was detected.

What each configuration adds

Route	Components	Operational obligations
R0	PostgreSQL; one Gemini call; four existing financial tools	Maintain existing typed tools and accounting gates.
R1	PostgreSQL; Gemini planner and answer calls; broader typed analytics	Maintain query compiler, calculation provenance and tool schemas.
R2	R1 plus PostgreSQL lexical document search	Ingest/version authorized documents; maintain lexical retrieval.
R3	Broad analytics plus Gemini managed File Search and final answer generation	Maintain managed store permissions, indexing and citation-to-document resolution; raw ranks unavailable.
R4	R2 plus Gemini embeddings and PostgreSQL pgvector hybrid retrieval	Maintain embedding versions, vector index, workspace filters and fusion.
R5	R4 plus Jev reranking	Operate another provider credential, candidate-ranking contract and limits.
R6	R4 retrieval with Wren analytics replacing the custom query compiler	Operate the pinned engine, semantic model and SQL provenance restrictions.
R7	R4 plus separate Gemini reviewer call using the same model weights	Maintain review prompts and evidence contract; correlated errors remain possible.
R8	R4 plus Jev reviewer	Maintain calibrated development threshold, reviewer schema and second-provider limits.

These are implementation responsibilities, not an invented complexity score. The simpler existing product remains the default when comparative benefit is below target or unresolved.

Observed tradeoff view	Not dominated on measured point estimates
All six families	R0, R1, R2, R3, R4, R5, R6
Financial + analytics	R0, R1, R2, R3, R4, R5, R6
Documents + scenario	R0, R1, R2, R3, R4, R5, R6, R8

Dominance compares overall and targeted success, execution-error rate, p95 harness latency and tracked cost per successful task. It ignores interval uncertainty and does not establish statistical superiority or a production winner.

Limitations and interpretation

1. All inputs are synthetic. References are independently implemented decimal fixture calculations, not audited business records. No independent human adjudication is claimed.

2. The primary score measures structured task success against frozen facts, scope, evidence IDs, proposal lines and explicit assumption/clarification tags. Natural-language summary faithfulness and usefulness still need human review.

3. Grounding scores measure source provenance: required source IDs, actual result-value bindings and deterministic calculations. They do not establish full semantic entailment of every citation. Supplemental claims without an independent reference or exact trusted-result binding are penalized as unassessed.

4. Automated scores are out of 90. The 10-point human clarity dimension and the combined 100-point score are unavailable; they are not imputed by another model.

5. Provider/tool errors count as failed end-to-end episodes. Conditional quality excludes status:error episodes and includes intentional blocks; it is not a replacement denominator.

6. Confidence intervals resample scenario groups within families. Paraphrases and repeat runs are correlated and stay together. Small synthetic samples do not establish real-world accuracy or safety.

7. All displayed 95% bootstrap intervals are unadjusted for the nine predeclared component contrasts and additional descriptive contrasts. They are exploratory effect estimates, not simultaneous confidence guarantees or claims of statistical significance. Two-group pilot and five-group-per-family held-out intervals may have poor coverage; zero-width empirical intervals do not establish certainty.

8. Reported monetary amounts are paid-equivalent estimates from returned usage and configured rates, not invoices. Unreported usage remains a separately identified conservative reservation; indexing/setup is separate.

9. Managed File Search uses the Interactions API to identify source citations. Its citation order is not a raw retrieval rank or exposed passage list. Authorized source IDs are mapped to the shared canonical documents before answer generation. This is managed source selection, not a like-for-like chunk retriever; raw-passage recall, MRR and nDCG are unavailable unless independently exposed.

10. Jev/LLM reviewers see supplied evidence, not ground truth, and can share generator errors. Their approval cannot override deterministic validation or database permissions.

11. The Wren route evaluates the pinned OSS semantic engine with the shared planner and approved views. It does not measure the complete commercial Wren product.

12. An incomplete route has no full-study conclusion. Missing episodes are reported separately, never silently removed or treated as free successful requests.

13. Episode completeness and snapshot integrity are separate gates. Changed or unverified before/after data makes this report ineligible for comparative selection even if every episode is present. Individual routes with hard-gate violations remain disqualified.

14. Queue-transport wall latency includes local dispatch delays and is not production request latency. Provider durations are measured separately.

15. All study generators normalized to Gemini3.5FlashLite; R0 is an architecture reproduction. Same-model critic errors may correlate.

16. Synthetic stress cases have 50% flawed prevalence; these results do not estimate production precision.

17. Correct/flawed pairs share source scenarios. Reused templates and fault patterns reduce independence; no publication-grade statistical power is claimed.

18. The deterministic baseline checks schema, citation existence and exact source amount/currency pairs. It intentionally cannot detect semantic period/entity/narrative contradictions.

19. Each semantic reviewer is run on every candidate, including code-detectable faults. Combined pipeline metrics show their additional contribution after code checks.

20. Provider errors are separate from semantic decisions. The operational column blocks on errors; the conditional column includes only completed reviewer responses.

21. Jev thresholds are selected only on development cases and frozen before holdout. Maximum hazard is an uncalibrated risk score; overall Brier calibration is unavailable.

22. LLM critic and main generator use the same Gemini3.5FlashLite weights in separate contexts; their errors are not statistically independent.

23. Reference fault labels are synthetic authored cases, not independently adjudicated human judgments. Narrative flaws are measured by these labels, not a judge-model truth claim.

Provenance and reproduction

All figures are vector graphics generated from the explicit saved JSON inputs. The exporter performs no provider requests, database queries, deployments or score recomputation. Source result values are retained; missing measurements remain N/A.

Exporter: daybook-pdf-report-v2. ReportLab: 5.1.0. Input bundle SHA-256: 5667ad2cbb79b2a3e38151e1f09cf9d8d69379a7f02dd2293ccadae0acd88cae

main results: results.json

SHA-256: eeb43bbad42e974dcb3c6fbc030f314304be49130b475313d8e785613da1ee29

verifier results: holdout-results.json

SHA-256: 6128cf2db38261603a56d895c84e10ac6faca56b8a7241fd8d740b88b835f4f6

retrieval supplement: retrieval-and-failures.json

SHA-256: a3665accded64cbed2e321fc8b8326219c4dd56c1143e89249a2b4b49a2fd234

common-capability analysis: common-capabilities.json

SHA-256: bef125c18ca24e5f44ff7656184181e6fa2b639c2dab380b1dedf408f16cbc76

common-capability analysis provenance: analysis-provenance.json

SHA-256: 8bdc48162b698ccad3dcbf908e2f7362c6f6753c2b8136fb647933db13cdba9a

weight-sensitivity analysis: sensitivity.json

SHA-256: 026bccae461c027e314040376f33ecdd90d26bb2cafad7982fa7a9495b0f52c0

weight-sensitivity analysis provenance: analysis-provenance.json

SHA-256: 4f612c88c7b35fedfe7025b681bee74e2fa4f2ef5fdfa0f72bde75411675c5b4

decision analysis: decisions.json

SHA-256: 3b44c1478dcd048e918d36bda0b85d27be940a201e15c261f02044c9a16cd5eb

decision analysis provenance: analysis-provenance.json

SHA-256: c1025d2f6408bf013af9da27a7fe36d29be462e346de82bf5fb59e91b237112f

The accompanying provenance JSON binds these inputs, the exporter source, validation results and output PDF hash. Footer hashes identify this snapshot on every page.

Original result source revision: 79dbdc9b2eddd9953815cebfc0f01a9d22c37f10. The working-tree/source manifest in the run directory identifies actual experimental code; a published ancestor alone is not sufficient.

Reproduce this export

```
.tools/report-render/bin/python scripts/export-research-report.py --main-results docs/evidence/research-runs/holdout-v1/results.json
--verifier-results docs/evidence/research-verifiers/verifiers-v1/holdout-results.json --verifier-split holdout --output
docs/evidence/research-runs/holdout-v1/study-report.pdf --supplement
docs/evidence/research-runs/holdout-v1/supplemental/retrieval-and-failures.json --common
docs/evidence/research-runs/holdout-v1/common-capabilities/common-capabilities.json --sensitivity
docs/evidence/research-runs/holdout-v1/sensitivity/sensitivity.json --decisions
docs/evidence/research-runs/holdout-v1/decisions/decisions.json
```

Renderer references: ReportLab Platypus and Graphics user guides (docs.reportlab.com/reportlab/userguide/ch5_platypus/ and ch11_graphics/). Scientific interpretation and experimental limitations come from the input study artifacts.